



Original research

Acute physiological responses with varying load or time under tension during a squat exercise: A randomized cross-over design



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ABSTRACT

Objectives: Despite advancements in resistance training (RT) methods, the acute metabolic and neuromuscular responses to time under tension (TUT) and load remains poorly understood. The aim of the present study was to investigate how TUT or resistance load impact muscular activation and blood lactate during an RT session.

Design: A randomized cross-over design.

Methods: Participants performed a squat exercise in three different conditions: baseline protocol (B_{PRO}; three sets of eight repetitions with four second repetitions at 60% of 1RM) long duration protocol (LD_{PRO}; six second repetitions) and high load protocol (HL_{PRO}; 70% 1RM).

Results: Muscular activation of the vastus lateralis and biceps femoris, and blood lactate were assessed. Blood lactate was ~19% and ~26% higher after set one and three in LD_{PRO} compared to B_{PRO} ($P \leq 0.011$) and ~17% higher for LD_{PRO} compared to HL_{PRO} ($P = 0.002$). Additionally, blood lactate was ~17% higher for HL_{PRO} compared to B_{PRO} after the third set of exercise ($P = 0.003$). Vastus lateralis activation was ~10% higher for HL_{PRO} compared to B_{PRO} and LD_{PRO} for sets one and two. Biceps femoris activation was ~17% higher for HL_{PRO} compared to B_{PRO} for set one ($P = 0.023$) while for set two HL_{PRO} was greater than B_{PRO} and LD_{PRO} (~19% and ~14%, respectively; $P \leq 0.007$).

Conclusions: Squatting with higher TUT caused a greater impact on the metabolic responses than lower TUT or higher loads, whereas an increase in training load resulted in greater muscle activation than higher TUT or lower training load.

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Practical implications

- The manipulation of time under tension and the training load (percent one-repetition maximum) during a resistance training session results in different acute physiological responses.
- An increase in the time under tension during the Smith machine squat exercise results in an increase in blood lactate, which may be linked to muscle hypertrophy in long-term training.

- An increase in training load during a Smith machine squat exercise results in greater muscle activation, and an increased blood lactate response was observed after a third set.
- While an increase in time under tension results in an overall increased metabolic response, greater training load augmented muscle activation and the metabolic response.

1. Introduction

Resistance training (RT) is an effective way for improving muscle strength, size and function. To achieve optimal RT outcomes (i.e., increased muscular hypertrophy, strength, or power), the manipulation of specific RT variables such as time under tension (TUT) and load [Percent one repetition maximum (1RM)] must be

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considered.¹ Understanding how to best manipulate these specific variables remains limited. TUT and load are known to promote different acute physiological responses,² and these responses may impact long-term adaptations such as increased strength or muscle size.^{3,4}

The increase in strength that occurs in the initial weeks with RT is often related to an increased capacity to activate the muscles involved in the task.^{5,6} When TUT or load are manipulated, changes in neuromuscular activation occur.^{7–9} Sakamoto and Sinclair⁹ simultaneously manipulated both TUT (exercise to failure) and load, which resulted in higher muscular activation when a lower TUT and higher load were used. Similarly, Schoenfeld et al.¹⁰ compared higher vs. lower load RT to failure, and found that higher loads and lower TUT resulted in higher muscle activation. Based upon the available literature higher loads result in greater neuromuscular activation, although none of these studies have investigated the effect without RT to muscle failure or controlling for TUT between conditions. Therefore, a consensus on the impact of TUT and load on acute muscular activation has not been reached. Performing exercise to muscle failure, which is not necessarily needed for desired adaptations,¹¹ may be contraindicated given people may present physical limitations (e.g., osteoarthritis) or may be in a reduced volume training phase. Additionally, a methodological issue is that studies have not equated the RT variables between the different protocols.¹² Therefore, further investigating muscular activation during acute RT sessions without task failure is warranted.

The increase in muscle size (hypertrophy) due to RT has been attributed to an augmented metabolic response.^{13,14} An increase in testosterone has long been thought as one possible factor contributing to muscle hypertrophy,¹⁵ although some literature has demonstrated that muscle hypertrophy can occur without changes in testosterone.¹⁶ An increase in mTORC1 signalling is another possible mechanism regulating muscle hypertrophy.¹³ An increase in blood lactate (Blood_{LAC}) contributes to testosterone production¹⁷ and mTORC1 signalling.¹³ However, changes in Blood_{LAC} concentrations with an acute RT session have resulted in mixed findings such as greater increases in Blood_{LAC} have been observed with higher^{18,19} and lower^{12,20} TUT protocols. It is unclear how the manipulation of TUT or load in isolation will affect Blood_{LAC} responses during a RT session.

Given that previous studies have not isolated a single RT variable, well controlled studies are warranted to better understand the impact of each variable on RT outcomes. Therefore, the aim of the present study was to investigate how two different protocols influence muscular activation [surface electromyography (EMG)] and metabolic responses (Blood_{LAC}) during a RT session. It was hypothesised that i) higher load RT would result in greater neuromuscular activation, and ii) higher TUT would result in a greater Blood_{LAC} response. The finds of the present study aim to provide important information regarding the impact of TUT and load on acute physiological responses to RT.

2. Materials and methods

Twenty resistance trained (RT ≥ six months at least three times a week), healthy (no traumatic lower-body injuries) males (age: 25 ± 4 years; height: 1.76 ± 0.06 m; body mass: 78 ± 9 kg; body fat: 13 ± 4%; 60% of 1RM: 76 ± 15.1 Kg; 70% of 1RM: 88 ± 17.7 Kg; 1RM: 127 ± 25.3 Kg) participated in this study. The study was approved by the local ethics committee. All participants provided written informed consent prior to their participation and this study was carried out in accordance to The Declaration of Helsinki. The funding organizations provided financial support but did not interfere with the research in any part of the process.

Table 1

Descriptive characteristics of the resistance exercise protocols.

	Protocol		
	B _{PRO}	LD _{PRO}	HL _{PRO}
Scheduled			
Sets × repetitions	3 × 8	3 × 8	3 × 8
Load (% 1RM)	60%	60%	70%
Duration of repetitions (CON:ECC) (s)	4 (2:2)	6 (3:3)	4 (2:2)
Total TUT	32 s	48 s	32 s
Rest between sets (min)	3	3	3
Range of motion of the knee (°)	80	80	80
Actually performed			
Duration of the repetition (s)	4.00 ± 0.04	5.97 ± 0.08*	4.00 ± 0.05
Range of motion of the knee (°)	81 ± 3	81 ± 3	80 ± 3
Load volume (sum of 8 repetitions, kg)	609 ± 121	609 ± 121	710 ± 141*

1RM = one repetition maximum; B_{PRO} = Baseline protocol; LD_{PRO} = Long duration protocol; HL_{PRO} = High load protocol. CON = Concentric; ECC = Eccentric; TUT = Time under tension.

* Significantly different than other protocols ($P < 0.001$). Data are (Means ± SD).

This study used a randomized crossover design to examine muscular activation and metabolic responses of RT protocols differentiated by the TUT [total set duration: 32 s (two seconds eccentric + two seconds concentric): vs. 48 s (three seconds eccentric + three seconds concentric)] or load (60% or 70% of 1RM). Participants were instructed to avoid any exercise for 24 h before each session, and to maintain the same routine regarding food intake and sleep during the period of study. During session one, participants were familiarized with the 1RM test by performing a 1RM Smith machine squat test. Next, one of the repetition durations and submaximal intensities were performed which was randomly assigned (four seconds TUT at 70% 1RM or six seconds TUT at 60% 1RM). In addition, body height, mass, and fat percentage [calculated using the equation of 7-skinfold (pectoralis, axilla, triceps, subscapular, abdominal, suprailiac and thigh)²¹] measurements were conducted. Session two consisted of a 1RM test followed by an additional familiarization period where the participants practiced performing the squat exercise with the other repetition duration that had not been practiced during session one. Following the two familiarization sessions, each participant performed a baseline protocol (B_{PRO}: three sets of eight repetitions with four second TUT for each repetition at 60% 1RM), a long duration protocol (LD_{PRO}: three sets of eight repetitions with six second TUT for each repetition at 60% 1RM), and a high load protocol (HL_{PRO}: three sets of eight repetitions with four second TUT for each repetition at 70% 1RM; Table 1) in a randomized order over sessions three to five.

A calibrated electrogoniometer (Noraxon, Scottsdale, AZ, USA) was fixed on the right knee-joint (on the knee-joint center) using double-sided adhesive tape and elastic bands. The electrogoniometer was used to determine knee displacement for all participants in the study for which the Smith machine squats were to be performed for each condition. The duration of each muscle action was comprised of the time spent between the maximum (knee flexion) and minimum (knee extension) angular positions. Thus eccentric duration corresponded to the period between the minimum and maximum angular positions while concentric (CON) duration corresponded to the maximum and minimum angular positions.²² The electrogoniometer raw data were converted into angular displacement data and filtered through a fourth order Butterworth low-pass filter with a cutoff frequency of 10 Hz.²²

Surface EMG protocols (Biovision, Wehrheim, Germany) were carried out following the SENIAM guidelines. After preparing the skin (shave and clean with alcohol), bipolar surface electrodes (Ag/AgCl) were placed parallel to the muscle fibers on the subjects' right vastus lateralis (VL) and biceps femoris (BF) muscles. After the electrodes were attached, a silk paper was used to register

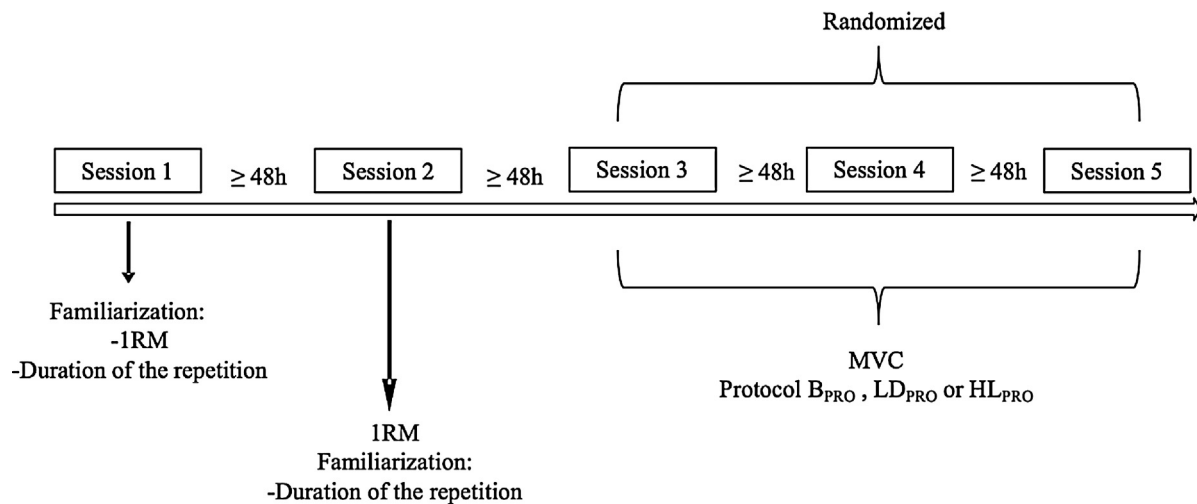


Fig. 1. Schematic diagram of the experimental design of the study. Where: 1RM = one repetition maximum; MVC = maximum isometric voluntary contraction; baseline protocol (B_{PRO}), longer duration protocol (LD_{PRO}) or higher load protocol (HL_{PRO}).

electrodes positions, taking into account bony extremities and relevant skin markings. These procedures were performed to ensure reproducibility of the electromyographic measurements between sessions. Data were collected with a sampling frequency of 2000 Hz. Data post-processing involved applying a second-order Butterworth band-pass filter (20–500 Hz) prior to a full-wave rectification and the area under the curve of the signal was then calculated.²³ The absolute area of the signal was determined for CON action and accounted for the different time durations.^{2,24} The area of the signal during the maximum isometric voluntary contraction (MVC; EMG_{MVC}) was measured as the area of the signal over a 1 s epoch at MVC peak force (500 ms either side).² A mean of the eight repetitions was determined for each set and then normalized by the EMG_{MVC}. DasyLab program (DasyLab 11.0; Measurement Computing Corporation, Norton, MA, USA) was used to record and analyse the data. The intraclass correlation coefficient (ICC^{3,1}) of the absolute EMG from CON muscle actions across sessions three to five was 0.97.

Blood samples were collected for all protocols during the rest period and one minute after the first and third sets. Blood was collected from a puncture to the subjects' left earlobe using sterile disposable lancets. The earlobe was cleaned with neutral soap and water and then sterilized with 70% alcohol before puncturing. A 30 µL sample of blood was collected into heparinized capillary tubes, which were transferred into other tubes containing 60 µL of 1% sodium fluoride and then stored in a refrigerator maintained at a temperature of -20 °C. Subsequently, the samples were thawed and analysed in duplicates on a Yellow Springs Sport 1500 Lactate Analyzer device (YSA, Inc., Yellow Springs, OH, USA) in order to measure Blood_{LAC}.²²

At the beginning of the first session, anthropometric measurements were performed. Participants were positioned in the Smith machine (Master, Brazil) with standardized bar position (on the shoulders), foot position and range of motion. The 1RM test instruction was performed as previously described in the literature.²⁵ The 1RM test (which was the same set up as main session) was used to determine the submaximal load for the rest of session one. After 10 min of rest, participants performed three sets at 50% 1RM with the different TUT protocols of four seconds (B_{PRO} and HL_{PRO}: two seconds eccentric + two seconds concentric) and six seconds (LD_{PRO}: three seconds eccentric + three seconds concentric) which were randomized between the two familiarization sessions (sessions one and two) in order to familiarize them on the different protocols in this study. A metronome was used to provide audi-

tory pacing for the desired repetition timing. During session two, the 1RM was re-evaluated and used to set the submaximal load for the subsequent training sessions. The ICC^{3,1} of the 1RM between familiarization and test sessions was 0.92 (Fig. 1).

During sessions three to five, participants performed three MVCs, at 85° of knee-flexion. After a rest period of 10 min, participants performed B_{PRO}, LD_{PRO} or HL_{PRO} (Table 1). Blood samples were taken one minute after set one and three, for all protocols.

Data are reported as mean ± standard deviation (SD). Variables were tested for normality and homogeneity of the variances using Shapiro-Wilks and Levene's tests, respectively. A paired *t*-test was used to compare volume load (sum of eight repetitions) between load (60% of 1RM vs. 70% of 1RM). Repeated-measures one-way ANOVA tests were used to analyse the differences between protocols for range of motion and TUT. Repeated-measures two-way ANOVA tests were used to analyse the differences between protocols for normalized EMG with a 3 × 3 ANOVA (protocol × set) and Blood_{LAC} 3 × 2 ANOVA (protocol × time). *Post hoc* comparisons were conducted using Bonferroni *post hoc* tests. Cohen's *d* effect sizes (ES) were calculated and classified as follows: <0.20 = "trivial"; 0.20–0.49 = "small"; 0.50–0.79 = "moderate" or >0.80 = "large". Statistical analyses were performed using SPSS version 24 (IBM Corporation, Armonk, New York, USA), with $\alpha = 0.05$.

3. Results

Protocols did not present differences in range of motion (one-way ANOVA, $F_{2,38} = 1.122$, $P = 0.336$; Table 1). However, as expected, TUT was different between protocols (one-way ANOVA, $F_{2,38} = 12.268$, $P < 0.001$) with LD_{PRO} being higher than B_{PRO} and HL_{PRO} (Bonferroni, $P < 0.001$; ES ≥ 29.73 'large'; see Table 1).

There was an interaction effect for Blood_{LAC} as well as main effects for protocol and time (two-way ANOVA, $F_{2,38} \leq 15.427$, $P < 0.001$). Blood_{LAC} was similar between protocols at rest but different after set one and three (Fig. 2). LD_{PRO} presented a ~19% increase in Blood_{LAC} after set one (Bonferroni, $P = 0.011$; ES = 0.64 'moderate') and ~26% after the third set compared to B_{PRO} (Bonferroni, $P < 0.001$; ES = 0.95 'large'). After the third set, LD_{PRO} also showed a ~8% increase in Blood_{LAC} compared to HL_{PRO} (Bonferroni, $P = 0.002$; ES = 0.30 'small') while HL_{PRO} presented a ~17% increase in Blood_{LAC} compared to B_{PRO} (Bonferroni, $P = 0.003$; ES = 0.43 'small'; Fig. 2.).

Normalized muscular activation for both VL and BF muscles presented main effects for protocol and time (two-way ANOVA, $F_{2,38} \geq 4.607$, $P \leq 0.016$) but only VL presented an interaction (two-

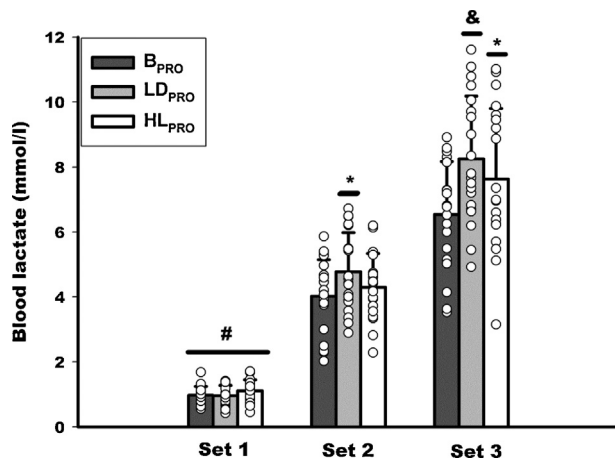


Fig. 2. Blood lactate at rest, after the first and third set for baseline protocol (B_{PRO}), longer duration protocol (LD_{PRO}) or higher load protocol (HL_{PRO}). Differences between protocols were detected (two-way ANOVA, $P < 0.001$). Symbols indicate: # all protocols were lower at rest compared to the other conditions, * higher than B_{PRO} at the same set, § higher than B_{PRO} and HL_{PRO} at the same set. Open circles indicate individual values. Data are mean $\pm$ SD ($n = 20$).

way ANOVA, $F_{4,76} = 3.384$, $P = 0.013$). HL_{PRO} led to $\sim 10\%$ increase in VL muscular activation compared to B_{PRO} and LD_{PRO} for the first [Bonferroni, $P \leq 0.016$; ES ≤ 0.55 'moderate' and $\sim 13\%$ and $\sim 11\%$

(respectively)] and second sets (Bonferroni, $P \leq 0.011$; ES ≤ 0.53 'moderate'; Fig. 3.). Similarly, BF muscle activation was higher for HL_{PRO} ($\sim 17\%$) compared to B_{PRO} at set one (Bonferroni, $P = 0.023$; ES = 0.61 'moderate'), while for set two HL_{PRO} presented greater muscle activation than B_{PRO} ($\sim 19\%$; Bonferroni, $P < 0.001$; ES = 0.61 'moderate') and LD_{PRO} ($\sim 14\%$; Bonferroni, $P = 0.007$; ES = 0.37 'small'). Moreover, BF activation for LD_{PRO} was 6% higher than B_{PRO} for set two (Bonferroni, $P = 0.011$; ES = 0.17 'trivial').

4. Discussion

To the authors knowledge this is the first study to investigate how the manipulation of load (HL_{PRO}) or TUT (LD_{PRO}), impact muscular activation and Blood_{LAC} compared to a baseline RT session (B_{PRO}). A session with greater TUT (LD_{PRO}: Total TUT = 48 s; Load = 60% 1RM), resulted in a greater lactate response compared to a baseline (B_{PRO}: Total TUT = 32 s; Load = 60% 1RM) or higher load (HL_{PRO}: Total TUT = 32 s; Load = 70% 1RM). However, HL_{PRO} demonstrated higher muscular activation compared to a B_{PRO} and LD_{PRO}. Overall, the results support our hypothesis that RT with higher TUT (LD_{PRO}) or load (HL_{PRO}) would provide preferential increases in specific metabolic responses compared to B_{PRO}. When protocols are matched for RT variables (set, repetitions, TUT, load, rest duration, and range of motion) with only TUT or load being adjusted, specific metabolic and neuromuscular responses ensue.

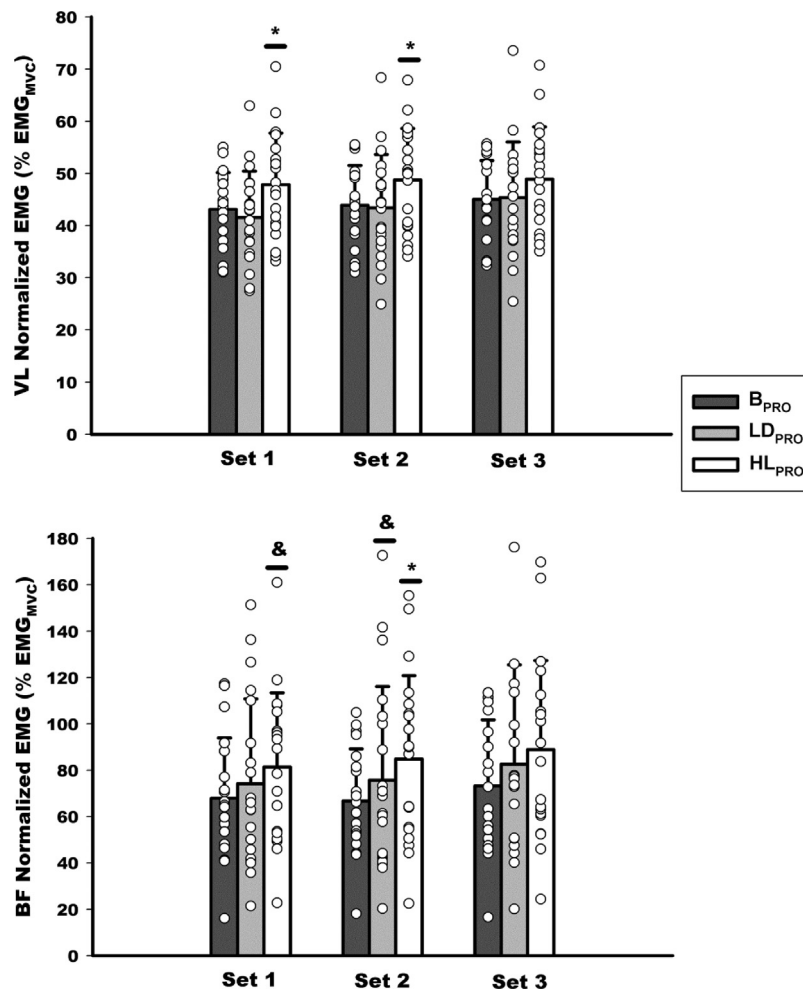


Fig. 3. Mean normalized muscular activation (% of EMG during MVC; EMG_{MVC}) across eight repetitions the three sets of the squat exercise for (A) vastus lateralis (VL) and (B) biceps femoris (BF) for baseline protocol (B_{PRO}), longer duration protocol (LD_{PRO}) or higher load protocol (HL_{PRO}). Differences between protocols were detected in both muscles (two-way ANOVA, $P \leq 0.018$). Symbols indicate: § higher than B_{PRO} at the same set; * higher than B_{PRO} and LD_{PRO} at the same set. Open circles indicate individual values. Data are mean $\pm$ SD ($n = 20$).

An increase in TUT from 4 s to 6 s per repetition can promote an acute increase in Blood_{LAC} (22%). Additionally, the increase in load by 10% (60%–70% 1RM) also significantly increased Blood_{LAC} by ~17% compared to baseline but only after the third set, which may indicate that an increased training load also impacts the metabolic response as the RT session progresses. However, within our design, increasing Blood_{LAC} appears to be more influenced by a change in TUT than load. Previous studies have reported contradictory results with an increase in Blood_{LAC} when longer^{19,22,26} or shorter^{12,20} TUT was used. Discrepancies between studies may be due to methodological differences. For instance, all previous studies manipulated at least two different RT variables when comparing protocols, making it difficult to link the specific RT variable to the physiological outcome. Given the present study only manipulated one variable within a condition (TUT or load), and controlled the other variables (set, repetition, rest, range of motion), our results provide novel insight into the impact TUT and load have on Blood_{LAC}.

An increase in Blood_{LAC} was previously showed to be due to different physiological factors such as type II fiber type recruitment, lactate removal, O₂ dependent metabolism and/or accelerating glycolysis.^{27,28} Although, these factors may help to explain the increases in Blood_{LAC} in the present study, during RT exercise the lactate removal may be the primary factor explaining the changes in Blood_{LAC}.^{18,26} The amount of lactate found in the blood is directly proportional to the difference between lactate influx and efflux to the bloodstream. Higher Blood_{LAC} is expected when the lactate influx in the bloodstream is higher than efflux.^{18,26} When an exercise is performed with a lower load but for a long period of time, even with a lower rate of lactate production, the longer TUT may lead to a higher blood lactate which may explain the results of the present study. Moreover, the present investigation showed an increase in ~17% of Blood_{LAC} (but small effect size) which occurred only after the third set with HL_{PRO} compared to B_{PRO}. This may suggest that more sets could result in Blood_{LAC} increases with a higher load, regardless of TUT. Given that higher muscle activation may lead to more muscle involvement, an increased metabolic response may occur resulting from greater muscle involvement.²⁹ Although muscle fiber types were not measured in this study, it is expected that a greater recruitment of Type II fibers would be observed with HL_{PRO}, which is known to increase Blood_{LAC}.^{12,20} The superior glycolytic power of Type II fibers compared to Type I fibers³⁰ may increase lactate concentration which could explain the differences between HL_{PRO} and LD_{PRO} after the third set. However, it is important to highlight that the increase in load was not enough to provide a higher metabolic response than TUT. These data may have implication for long-term exposition (e.g., muscle hypertrophy) considering the association between higher Blood_{LAC} and testosterone.¹⁷

Increasing the RT load from 60% to 70% 1RM with HL_{PRO} led to an increase of ~14% in normalized muscular activation across the exercise session compared to B_{PRO} for the BF muscle and both B_{PRO} and LD_{PRO} for the VL muscle. Similarly, other studies comparing the impact of different intensities on muscular activation with different TUT^{7–9} showed increased muscular activation when higher loads were used. Although, each of these previous investigations manipulated two or more RT variables and therefore did not isolate TUT or load. The present study demonstrated that increasing the load of the exercise, provides an increase in muscular activation for both VL and BF muscles.

Acute increases in muscular activation are known to be associated with changes in the number of motor units (MU) being recruited, modulation of discharge frequency and synchronization of motor units.⁵ Increased synchronization between action potentials has also been observed with higher load exercise.³¹ This increased synchronization is directly reflected in an augmented EMG amplitude.³² It is possible that the increase in muscular activation

found with HL_{PRO} is due to an increase in MU recruitment and synchronization of motor units. Although LD_{PRO} also led to greater muscular activation compared to the B_{PRO}, the difference was small (6%) with a trivial effect size. These data suggest that load rather than TUT appears to influence a change in acute measures of increased muscular activation.

5. Conclusion

Squat exercise performed in the Smith machine with greater TUT increased Blood_{LAC} compared to B_{PRO} and HL_{PRO}, whereas an increase in training load resulted in a greater EMG response in the knee extensors compared to B_{PRO} and LD_{PRO}. These data have implications for RT program design to enhance performance outcomes for athletes and special population.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jsams.2020.07.015>.

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